The dubbing flow

How a YouTube link becomes a dubbed MP4 and a Roman Urdu transcript.

01 **Submit the link**

The browser sends a YouTube link to the Flask app.

↓

02 **Import the video**

yt-dlp downloads the source video for processing.

↓

03 **Transcribe the speech**

Faster Whisper produces English text with timestamps.

↓

04 **Translate each segment**

Qwen runs through Ollama and returns two forms of Urdu.

AUDIO PATH

Urdu script

Piper generates Pakistani Urdu speech.
FFmpeg syncs it with the original video.

Dubbed video / MP4

TEXT PATH

Roman Urdu

The app writes each translated line
with its timestamp.

Transcript / TXT

Review a line before applying it

Edit the Roman Urdu line, preview its audio, then apply the change.

URDU DUB STUDIO / POSTER 02

The Docker setup

How Docker Compose starts Urdu Dub Studio on your Surface.

01

ollama

Starts the local model server on the Compose network.

Health check passes

02

model-setup

Pulls qwen3.5:4b into Ollama and exits after setup.

Exits successfully (code 0)

03

app

Starts Flask and connects to Ollama at ollama:11434.

App health check passes

OPEN IN YOUR SURFACE BROWSER

http://localhost:8765

What stays on disk

Docker volumes

ollama_models keeps the Qwen model.
speech_models caches speech models.

Temporary app storage

Download finished videos and transcripts from the browser.